

1. IAM SERVICE ACCOUNT & CONTROLLER CONFIGURATION

File: Screenshot 2026-05-19 at 1.21.59 AM.jpg

Shows the setup and creation of the IAM Service Account (sa) mapped to the AmazonEKSLoadBalancerControllerRole, along with the initial rollout restart command.

```
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--approve
2026-05-19 08:18:54 [i] 1 existing iamserviceaccount(s) (kube-system/aws-load-balancer-controller) will be excluded
2026-05-19 08:18:54 [i] 1 iamserviceaccount (kube-system/aws-load-balancer-controller) was excluded (based on the include/exclude rules)
2026-05-19 08:18:54 [i] serviceaccounts that exist in Kubernetes will be excluded, use --override-existing-serviceaccounts to override
2026-05-19 08:18:54 [i] no tasks
jatinrai@Jatins-MacBook-Air downloads % eksctl create iamserviceaccount \
--cluster demo-cluster \
--region us-east-1 \
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--approve \
--override-existing-serviceaccounts to override
2026-05-19 08:11:42 [w] only one argument is allowed to be used as a name
jatinrai@Jatins-MacBook-Air downloads % eksctl create iamserviceaccount \
--cluster demo-cluster \
--region us-east-1 \
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--override-existing-serviceaccounts \
--approve
2026-05-19 08:12:09 [i] 1 existing iamserviceaccount(s) (kube-system/aws-load-balancer-controller) will be excluded
2026-05-19 08:12:09 [i] 1 iamserviceaccount (kube-system/aws-load-balancer-controller) was excluded (based on the include/exclude rules)
2026-05-19 08:12:09 [i] metadata of serviceaccounts that exist in Kubernetes will be updated, as --override-existing-serviceaccounts was set
2026-05-19 08:12:09 [i] no tasks
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1219332"
  uid: e9c3ee0e-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl annotate serviceaccount aws-load-balancer-controller \
-n kube-system \
eks.amazonaws.com/role-arn=arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole \
--overwrite
serviceaccount/aws-load-balancer-controller annotated
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1224339"
  uid: e9c3ee0e-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl rollout restart deployment aws-load-balancer-controller -n kube-system
deployment.apps/aws-load-balancer-controller restarted
jatinrai@Jatins-MacBook-Air downloads % kubectl get ingress -n game-2048
NAME          CLASS  HOSTS          ADDRESS          PORTS  AGE
ingress-2048  alb    *             k8s-game2048-ingress2-bcac0b5b37-682643449.us-east-1.elb.amazonaws.com  80     2d23h
jatinrai@Jatins-MacBook-Air downloads %
```



Low Battery

Your Mac will sleep soon unless plugged into a power outlet.

2. PODS RUNNING

File: Screenshot 2026-05-19 at 1.23.39 AM.jpg

Displays the execution of 'kubectl get pods -n kube-system' confirming that both pods for the aws-load-balancer-controller, coredns, and metrics-server are up and fully Running.

```
--cluster demo-cluster \
--region us-east-1 \
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--approve \
> --override-existing-serviceaccounts to override
2026-05-19 00:12:09 [!] only one argument is allowed to be used as a name
jatinrai@Jatins-MacBook-Air downloads % eksctl create iamserviceaccount \
--cluster demo-cluster \
--region us-east-1 \
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--override-existing-serviceaccounts \
--approve
2026-05-19 00:12:09 [i] 1 existing iamserviceaccount(s) (kube-system/aws-load-balancer-controller) will be excluded
2026-05-19 00:12:09 [i] 1 iamserviceaccount (kube-system/aws-load-balancer-controller) was excluded (based on the include/exclude rules)
2026-05-19 00:12:09 [i] metadata of serviceaccounts that exist in Kubernetes will be updated, as --override-existing-serviceaccounts was set
2026-05-19 00:12:09 [i] no tasks
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1219332"
  uid: e9c3eeeb-e9f7-4b45-baad-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl annotate serviceaccount aws-load-balancer-controller \
-n kube-system \
eks.amazonaws.com/role-arn=arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole \
--overwrite
serviceaccount/aws-load-balancer-controller annotated
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1224338"
  uid: e9c3eeeb-e9f7-4b45-baad-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl rollout restart deployment aws-load-balancer-controller -n kube-system
deployment.apps/aws-load-balancer-controller restarted
jatinrai@Jatins-MacBook-Air downloads % kubectl get ingress -n game-2048
NAME          CLASS    HOSTS          ADDRESS          PORTS    AGE
ingress-2048  alb     *             k8s-game2048-ingress2-bcac0b5b37-602643449.us-east-1.elb.amazonaws.com  80      2d23h
jatinrai@Jatins-MacBook-Air downloads % kubectl get pods -n kube-system
zsh: command not found: kubectl
jatinrai@Jatins-MacBook-Air downloads % kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-load-balancer-controller-4b48b86f79-mrb9c  1/1     Running   0           68s
aws-load-balancer-controller-4b48b86f79-rdsrn  1/1     Running   0           69s
coredns-fdc4b5f67-kxflf                1/1     Running   0           3d2h
coredns-fdc4b5f67-w7g7q                1/1     Running   0           3d2h
metrics-server-6f49c4bc6c-p4ts9         1/1     Running   0           3d2h
metrics-server-6f49c4bc6c-t7ggn        1/1     Running   0           3d2h
jatinrai@Jatins-MacBook-Air downloads %
```



Low Battery

Your Mac will sleep soon unless plugged into a power outlet.

3. INGRESS OUTPUT

File: Screenshot 2026-05-19 at 1.23.55 AM.jpg

Displays the execution of 'kubectl get deploy -n kube-system' verifying that desired deployment scales match the actual ready replicas for all core components.

```
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--approve \
--override-existing-serviceaccounts to override
2026-05-19 00:11:42 [M] only one argument is allowed to be used as a name
jatinrai@Jatins-MacBook-Air downloads % eksctl create iamserviceaccount \
--cluster demo-cluster \
--region us-east-1 \
--namespace kube-system \
--name aws-load-balancer-controller \
--role-name AmazonEKSLoadBalancerControllerRole \
--attach-policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
--override-existing-serviceaccounts \
--approve
2026-05-19 00:12:09 [i] 1 existing iamserviceaccount(s) (kube-system/aws-load-balancer-controller) will be excluded
2026-05-19 00:12:09 [i] 1 iamserviceaccount (kube-system/aws-load-balancer-controller) was excluded (based on the include/exclude rules)
2026-05-19 00:12:09 [i] metadata of serviceaccounts that exist in Kubernetes will be updated, as --override-existing-serviceaccounts was set
2026-05-19 00:12:09 [i] no tasks
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1219332"
  uid: e9c3ee0e-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl annotate serviceaccount aws-load-balancer-controller \
-n kube-system \
eks.amazonaws.com/role-arn=arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole \
--overwrite
serviceaccount/aws-load-balancer-controller annotated
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1224330"
  uid: e9c3ee0e-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl rollout restart deployment aws-load-balancer-controller -n kube-system
deployment.apps/aws-load-balancer-controller restarted
jatinrai@Jatins-MacBook-Air downloads % kubectl get ingress -n game-2048
NAME          CLASS    HOSTS          ADDRESS          PORTS    AGE
ingress-2048  alb     *             k8s-game2048-ingress2-bcac0b5b37-682643449.us-east-1.elb.amazonaws.com  80      2d23h
jatinrai@Jatins-MacBook-Air downloads % kubectl get pods -n kube-system
zsh: command not found: kubectl
jatinrai@Jatins-MacBook-Air downloads % kubectl get pods -n kube-system
NAME                                READY    STATUS    RESTARTS    AGE
aws-load-balancer-controller-6b48b86f79-mrb9c  1/1      Running   0           68m
aws-load-balancer-controller-6b48b86f79-rdsrn  1/1      Running   0           69m
coredns-fdc4b5f67-kxflf                 1/1      Running   0           3d2h
coredns-fdc4b5f67-w7q7g                 1/1      Running   0           3d2h
metrics-server-6f49c4bc6c-p6ts9          1/1      Running   0           3d2h
metrics-server-6f49c4bc6c-t7qgn          1/1      Running   0           3d2h
jatinrai@Jatins-MacBook-Air downloads % kubectl get deploy -n kube-system
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
aws-load-balancer-controller         2/2      2             2            3d
coredns                              2/2      2             2            3d3h
metrics-server                       2/2      2             2            3d2h
jatinrai@Jatins-MacBook-Air downloads %
```



Low Battery

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4. EKS NODE FLEET & SUBNET ALLOCATION

File: Screenshot 2026-05-19 at 2.50.07 PM.jpg

Displays 'kubectl get nodes', mapping out all active serverless AWS Fargate IP tracks currently operating inside the cluster data-plane.

```
--approve
2026-05-19 00:12:09 [i] 1 existing iamServiceAccount(s) (kube-system/aws-load-balancer-controller) will be excluded
2026-05-19 00:12:09 [i] 1 iamServiceAccount (kube-system/aws-load-balancer-controller) was excluded (based on the include/exclude rules)
2026-05-19 00:12:09 [i] metadata of serviceaccounts that exist in Kubernetes will be updated, as --override-existing-serviceaccounts was set
2026-05-19 00:12:09 [i] no tasks
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1219332"
  uid: e9c3eee0-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl annotate serviceaccount aws-load-balancer-controller \
-n kube-system \
eks.amazonaws.com/role-arn=arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole \
--overwrite
serviceaccount/aws-load-balancer-controller annotated
jatinrai@Jatins-MacBook-Air downloads % kubectl get sa aws-load-balancer-controller -n kube-system -o yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::570538546859:role/AmazonEKSLoadBalancerControllerRole
  creationTimestamp: "2026-05-18T18:23:24Z"
  name: aws-load-balancer-controller
  namespace: kube-system
  resourceVersion: "1224330"
  uid: e9c3eee0-e9f7-4545-bae6-18d28290dec0
jatinrai@Jatins-MacBook-Air downloads % kubectl rollout restart deployment aws-load-balancer-controller -n kube-system
deployment.apps/aws-load-balancer-controller restarted
jatinrai@Jatins-MacBook-Air downloads % kubectl get ingress -n game-2048
NAME          CLASS    HOSTS          ADDRESS          PORTS    AGE
ingress-2048  alb     *             k8s-game2048-ingress2-bcac0b5b37-602643449.us-east-1.elb.amazonaws.com  80      2d23h
jatinrai@Jatins-MacBook-Air downloads % kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
aws-load-balancer-controller-6b48b86f79-mrb9c  1/1     Running   0           69m
aws-load-balancer-controller-6b48b86f79-rdsrn  1/1     Running   0           3d2h
coredns-fdc4b5f67-kxflf                    1/1     Running   0           3d2h
coredns-fdc4b5f67-w7g7g                    1/1     Running   0           3d2h
metrics-server-6f49c4bc6c-p4ts9             1/1     Running   0           3d2h
metrics-server-6f49c4bc6c-t7ggn             1/1     Running   0           3d2h
jatinrai@Jatins-MacBook-Air downloads % kubectl get deploy -n kube-system
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
aws-load-balancer-controller        2/2      2             2           3d
coredns                             2/2      2             2           3d3h
metrics-server                      2/2      2             2           3d2h
jatinrai@Jatins-MacBook-Air downloads % kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
fargate-ip-192-168-101-157.ec2.internal  Ready    <none>   3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-109-240.ec2.internal  Ready    <none>   3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-116-223.ec2.internal  Ready    <none>   3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-119-124.ec2.internal  Ready    <none>   3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-74-249.ec2.internal   Ready    <none>   14h   v1.34.7-eks-4136f65
fargate-ip-192-168-76-88.ec2.internal    Ready    <none>   3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-76-93.ec2.internal    Ready    <none>   3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-83-105.ec2.internal    Ready    <none>   3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-83-184.ec2.internal    Ready    <none>   14h   v1.34.7-eks-4136f65
fargate-ip-192-168-98-223.ec2.internal    Ready    <none>   3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-96-11.ec2.internal     Ready    <none>   3d15h  v1.34.7-eks-4136f65
```

5. EKS CLUSTER VERIFICATION

File: Screenshot 2026-05-19 at 2.50.38 PM.jpg

Shows 'eksctl get cluster --region us-east-1' identifying that 'demo-cluster' is completely active, verified, and active in AWS.

```
NAME                                READY    STATUS    RESTARTS   AGE
aws-load-balancer-controller-6b48b86f79-mrb9c  1/1      Running   0           68m
aws-load-balancer-controller-6b48b86f79-rdsrn  1/1      Running   0           69m
coredns-fdc4b5f67-kxf1f                 1/1      Running   0           3d2h
coredns-fdc4b5f67-w7g7q                 1/1      Running   0           3d2h
metrics-server-6f49c4bc6c-p4ts9          1/1      Running   0           3d2h
metrics-server-6f49c4bc6c-t7ggn          1/1      Running   0           3d2h
jatinrai@Jatins-MacBook-Air downloads % kubectl get deploy -n kube-system
NAME                                READY    UP-TO-DATE    AVAILABLE   AGE
aws-load-balancer-controller        2/2      2              2            3d
coredns                             2/2      2              2            3d3h
metrics-server                      2/2      2              2            3d2h
jatinrai@Jatins-MacBook-Air downloads % kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
fargate-ip-192-168-181-157.ec2.internal  Ready    <none>    3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-189-248.ec2.internal  Ready    <none>    3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-116-223.ec2.internal  Ready    <none>    3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-119-124.ec2.internal  Ready    <none>    3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-74-249.ec2.internal  Ready    <none>    14h    v1.34.7-eks-4136f65
fargate-ip-192-168-76-88.ec2.internal    Ready    <none>    3d14h  v1.34.7-eks-4136f65
fargate-ip-192-168-76-93.ec2.internal    Ready    <none>    3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-83-185.ec2.internal    Ready    <none>    3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-83-184.ec2.internal    Ready    <none>    14h    v1.34.7-eks-4136f65
fargate-ip-192-168-98-223.ec2.internal    Ready    <none>    3d15h  v1.34.7-eks-4136f65
fargate-ip-192-168-96-11.ec2.internal     Ready    <none>    3d15h  v1.34.7-eks-4136f65
jatinrai@Jatins-MacBook-Air downloads % eksctl get cluster
No clusters found
jatinrai@Jatins-MacBook-Air downloads % eksctl get clustereksctl get cluster --region us-east-1
Error: unknown resource type "clustereksctl"

Get resource(s)

Usage: eksctl get [flags]

Commands:
  eksctl get accessentry      Get access entry(ies)
  eksctl get addon            Get an Addon
  eksctl get capability       Get capability(ies)
  eksctl get cluster          Get cluster(s)
  eksctl get fargateprofile    Get Fargate profile(s)
  eksctl get iamidentitymapping Get IAM identity mapping(s)
  eksctl get iamserviceaccount Get iamserviceaccount(s)
  eksctl get identityprovider  Describe identity providers for cluster authentication and authorization
  eksctl get labels            Get labels for managed nodegroup
  eksctl get nodegroup         Get nodegroup(s)
  eksctl get podidentityassociation Get a pod identity association

Common flags:
  -C, --color string  toggle colorized logs (valid options: true, false, fabulous) (default 'true')
  -d, --dumpLogs      dump logs to disk on failure if set to true
  -h, --help           help for this command
  -v, --verbose int    set log level, use 0 to silence, 4 for debugging and 5 for debugging with AWS debug logging (default 3)

Use 'eksctl get [command] --help' for more information about a command.

For detailed docs go to https://eksctl.io/

Error: unknown resource type "clustereksctl"
jatinrai@Jatins-MacBook-Air downloads % eksctl get cluster --region us-east-1
NAME      REGION    EKSCTL CREATED
demo-cluster  us-east-1    True
jatinrai@Jatins-MacBook-Air downloads % █
```

6. ALB DNS WORKING

File: Screenshot 2026-05-19 at 1.24.09 AM.jpg

The real web browser output loading the '2048 game' perfectly via the public Application Load Balancer endpoint URL.

